# MILITARY SPECIFICATION

## BOLTS, AIRCRAFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bolts used in the manufacture of aircraft and aircraft accessories.

1.2 Classification. The bolts covered by this specification shall be of the types and sizes designated (see 6.2).

* 2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| QQ-A-200 | Aluminum Alloy Bar, Rod, Shapes, Structural Shapes, Tube and Wire, Extruded, General Specification for. |
| QQ-A-225 | Aluminum and Aluminum Alloy Bar, Rod, Wire, or Special Shapes, Rolled, Drawn, or Cold Finished, General Specification for. |
| QQ-A-430 | Aluminum Alloy Rod and Wire, for Rivets and Cold Heading. |
| QQ-P-416 | Plating, Cadmium (Electrodeposited). |
| QQ-S-781 | Strapping, Steel, and Seals. |
| PPP-H-1581 | Hardware (Fasteners and Related Items) Packaging of. |

Beneficial comments (recommendations, additions, deletions) and and pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5306

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MILITARY

| | | |
|---|---|---|
| | MIL-C-5541 | Chemical Conversion Coatings on Aluminum Alloys. |
| | MIL-S-5626 | Steel, Chrome-Molybdenum (4140) Bars, Rods, and Forging Stock (for Aircraft Application). |
| | MIL-S-6049 | Steel, Chrome-Nickel-Molybdenum (8740) Bars, Rods, and Forging Stock (for Aircraft Applications). |
| | MIL-S-6050 | Steel, Chrome-Nickel-Molybdenum (8630) Bars, Rods, and Forging Stock (for Aircraft Applications). |
| | MIL-H-6088 | Heat Treatment of Aluminum Alloys. |
| | MIL-S-6098 | Steel, Chrome Nickel-Molybdenum (8735) Bars, and Reforging Stock (Aircraft Quality). |
| | MIL-S-6758 | Steel, Chrome-Molybdenum (4130) Bars, and Reforging Stock (Aircraft Quality). |
| | MIL-I-6866 | Inspection, Penetrant Method of. |
| | MIL-I-6868 | Inspection Process, Magnetic Particle. |
| | MIL-H-6875 | Heat Treatment of Steels, Process for. |
| | MIL-S-7742 | Screw Threads, Standard, Optimum Selected Series, General Specification for. |
| | MIL-A-8625 | Anodic Coatings, for Aluminum and Aluminum Alloys. |
| * | MIL-S-8879 | Screws Threads, Controlled Radius Root with Increased Minor Diameter General Specification for. |
| | MIL-S-18732 | Steel Bars, Billets, Forgings, Tubing, (431) Special Quality. |

STANDARDS

FEDERAL

| | | |
|---|---|---|
| | FED-STD-151 | Metal, Test Methods. |

MILITARY

| | | |
|---|---|---|
| | MIL-STD-105 | Sampling Procedures and Tables for. |
| * | MIL-STD-129 | Marking for Shipment and Storage. |
| | MIL-STD-143 | Standards and Specifications, Order of Precedence for the Selection of. |
| | MIL-STD-147 | Palletized Unit Load. |
| | MIL-STD-410 | Nondestruction Testing Personnel Qualification And Certification (Eddy Current, Liquid Penetrant, Magnetic Particle, Radiographic And Ultrasonic) |
| | MIL-STD-414 | Sampling Procedures and Tables for Inspection by Variables for Percent Defective. |
| * | MIL-STD-794 | Parts and Equipment, Procedures for Packaging and Packing of. |
| * | MIL-STD-45662 | Calibration System Requirements. |
| | AN3 thru AN20 | Bolt - Machine, Aircraft. |
| | AN21 thru AN37 | Bolt and Clevis. |
| | AN42 thru AN49 | Bolt, Eye. |
| | AN173 thru AN186 | Bolt and Machine, Close Tolerance, Aircraft. |
| | AN315 | Nut, Plain, Airframe. |
| | AN320 | Nut, Plain, Castellated, Shear. |
| | MS20073 | Bolt, Machine, Aircraft, Drilled Head, Fine Thread. |
| | MS20074 | Bolt, Machine, Aircraft, Drilled Head, Coarse Thread. |
| | MS21083 | Nut, Self-Locking, Hexagon, Non-Metallic Insert, Low Height, 250°F. |



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* (Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

SOCIETY OF AUTOMOTIVE ENGINEERS AEROSPACE MATERIAL SPECIFICATION

AMS6300 Steel, .25 Mo (.35-.40C)

(Copies of SAE publications may be obtained from the Society of Automotive Engineers, Inc. 400 Commonwealth Drive, Warrendale, PA 15096.)

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D3951-82 Packaging,. Commercial

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material.

3.1.1 Non-corrosion-resistant steel. Non-corrosion-resistant steel bolts shall be manufactured from steel conforming to MIL-S-5626, MIL-S-6049, MIL-S-6050, MIL-S-6098, MIL-S-6758, or AMS 6300 (see 6.1.1).

3.1.2 Corrosion-resistant steel. Corrosion-resistant steel bolts shall be manufactured from material conforming to MIL-S-18732.

3.1.3 Aluminum alloy. Aluminum alloy bolts shall be manufactured from material conforming to QQ-A-225, of QQ-A-430.

3.1.4 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143.



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* 3.2 Heat treatment. Heat treatment of steel shall be in accordance with MIL-H-6875. Heat treatment of aluminum alloy shall be in accordance with MIL-H-6088.

3.2.1 Material conforming to MIL-S-18732 shall not be tempered in the range of 700° to 1,000° Farenheit (F).

3.3 Heads.

3.3.1 Bearing surface. The bearing surface of bolt heads shall be at right angles to the shank within 2 degrees for bolts 1/2 inch in diameter and less, and within 1 degree for bolts larger than 1/2 inch in diameter.

3.3.2 Eccentricity of head to shank. Eccentricity of head to shank shall be within a tolerance of 3 percent of the width across flats, or 3 percent of the diameter of the head in the case of clevis and eye bolts.

3.3.3 Top surface. The top surface of hexagon head bolts shall be flat and chamfered with the diameter of the top flat circle from 85 to 100 percent of the width across flats.

3.3.4 Depth of slots. The depth of slots in the heads of clevis bolts is measured from the highest part of the head to the intersection of the bottom of the slot with the head surface.

3.4 Straightness of shank. The straightness of shank shall be within the values specified in Table I when the bolt is rolled on a surface plate and the point of greatest deviation is measured with a feeler gage of 1/4 inch width.

TABLE I. Straightness of Shank.

| Bolt size | Deviation of bolt shank from plates., max (inches per inch of bolt length) |
|---|---|
| No. 10 and smaller | 0.0040 |
| 1/4, 5/16 | .0030 |
| 3/8, 7/16 | .0025 |
| 1/2 and larger | .0020 |

3.5 Physical properties.

3.5.1 Tensile strength. The tensile strength of hexagon head, eye, and clevis bolts shall be as specified in Table II when tested in accordance with 4.5.2.

3.5.2 Shear strength. The shear strength of all bolt types shall be as specified in Table II when tested in accordance with 4.5.3.



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3.5.3 Bend. Bolts shall withstand cold bending when tested in accordance with 4.5.4.

3.5.4 Hardness. All corrosion and noncorrosion-resistant steel bolts shall be within the Rockwell hardness range C-26 to C-32 when tested in accordance with 4.5.5.

3.6 Decarburization. Decarburization of noncorrosion-resistant steel bolts shall not exceed the limits shown in Table III, when tested in accordance with 4.5.6.

3.7 Head structure. A section of the head shall show no detrimental defects when inspected in accordance with 4.5.7.

3.8 Cracks. Bolts shall be free of cracks in any direction and location. A crack is defined as a clean crystalline break passing through the grain or grain boundary without the inclusion of foreign elements.

TABLE II. Strength Requirements.

| Size | Ultimate tensile strength (min) 1/ (pounds) | | | | Double shear strength (min) 2/ (pounds) | |
|---|---|---|---|---|---|---|
| | | Hexagon head bolts | | | All types of bolts | |
| | | Steel | | Al 3/ alloy | | |
| | Eyebolts steel | Fine 3/ thread | Coarse 3/ thread | fine thread | Steel | Al alloy |
| No. 6 | - | - | - | - | 2,120 | 1,080 |
| No. 8 | - | - | - | - | 3,000 | 1,570 |
| No. 10 | 1,150 | 2,210 | 1,800 | 1,100 | 4,250 | 2,092 |
| 1/4 | 2,450 | 4,080 | 3,360 | 2,030 | 7,360 | 3,650 |
| 5/16 | 3,910 (AN44) | 6,500 | 5,660 | 3,220 | 11,500 | 5,700 |
| 5/16 4/ | 5,290 (AN45) | 6,500 | 5,660 | 3,220 | 11,500 | 5,700 |
| 3/8 | 7,015 | 10,100 | 8,470 | 5,020 | 16,560 | 8,250 |
| 7/16 | 9,200 | 13,600 | 11,680 | 6,750 | 22,500 | 11,200 |
| 1/2 | 14,375 | 18,500 | 15,730 | 9,180 | 29,400 | 14,600 |
| 9/16 | 20,125 | 23,600 | 20,300 | 11,700 | 37,400 | 18,500 |
| 5/8 | - | 30,100 | 25,100 | 14,900 | 46,000 | 22,800 |
| 3/4 | - | 44,000 | 37,800 | 21,800 | 66,300 | 33,000 |
| 7/8 | - | 60,000 | - | 29,800 | 90,100 | 45,000 |
| 1 | - | 80,700 | - | 40,000 | 117,800 | 58,500 |
| 1-1/8 | - | 101,800 | - | 50,500 | 147,500 | 74,000 |
| 1-1/4 | - | 130,200 | - | 64,400 | 182,100 | 91,000 |

1/ The values shown for the ultimate tensile strength are for minimum values and are based on:

a. 125,000 psi for noncorrosion-resistant and corrosion-resistant steel.

b. 62,000 psi for aluminum alloy.



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The strength values shown for the eyebolts are based on the strength of the eye. The root area of the thread is the basis of calculation for the tensile strength of hexagon head bolts.

Clevis bolts shall have tensile strengths equal to one-half of the requirements for hexagon-head bolts when used with AN320 or MS21083 nuts. Clevis bolts are intended primarily for use in shear applications.

2/ Ultimate shear strengths are computed on the basis of 60 percent of the ultimate tensile strengths.

3/ Class of thread is as specified on the applicable standard drawing.

4/ Different from size 5/16 above in the design of the eye section.

TABLE III. Decarburization Limits.

| | Complete decarburization (inches) | Complete plus partial decarburization (inches) 1/ |
|---|---|---|
| Up to 3/8 incl | 0.004 | 0.010 |
| Over 3/8 to 1/2 incl | .005 | .012 |
| Over 1/2 to 5/8 incl | .006 | .014 |
| Over 5/8 to 3/4 incl | .006 | .017 |

1/ The permissible values shown are based on measurements of the entire periphery of the cross section and the average depth determined. The decarburization limits apply to the shank.

3.9 Seams. Bolts may possess seams, except in the locations noted hereinafter, which do not exceed the tolerances in Table IV. Seams extending through fillets or threads as shown in Figure 1 that exceed the limits shown in Table IV shall not be acceptable. Seams up to the depth indicated which have been rolled beneath the root of the threads shall not be cause for rejection.



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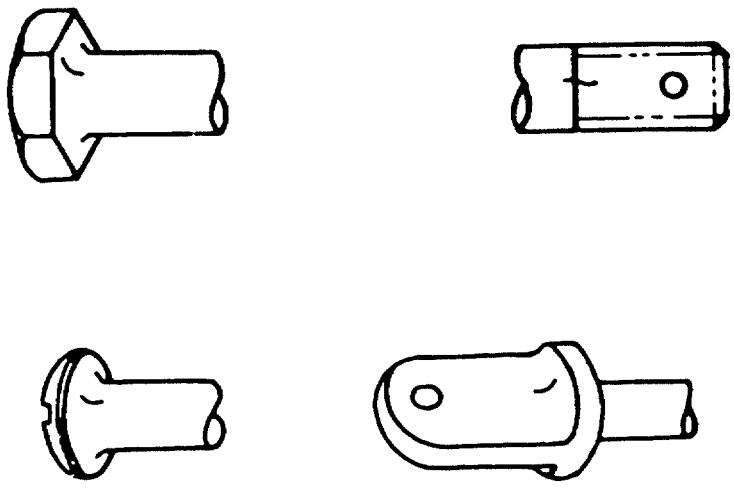


FIGURE 1. Seams extending through fillets or threads.



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TABLE IV. Limits for Seam Depth.

| Bolt size (inches) | 5/16 and under | 3/8 | 7/16 | 1/2 through 1-1/4 |
|---|---|---|---|---|
| Depth in shank (inches) | 0.005 | 0.006 | 0.007 | 0.008 |

3.10 Inclusions. Bolts possessing surface or sub-surface inclusions, as revealed by any of the methods noted herein, and not indicative of unsatisfactory quality, shall be accepted.

3.11 Defects on bolt heads. Bolt heads shall not possess more than three openings such as are described in 3.11.1.

3.11.1 Seams, inclusions, nicks, or gouges. Bolt heads shall not possess seams, inclusions, nicks or gouges on the following:

a. On the top of the bolt head exceeding twice the depth limits shown in Table IV, or

b. On the periphery of the bolt head exceeding twice the depth limits shown in Table IV for hexagon-head bolts or four times the depth limits shown in Table IV for bolts with other than hexagon heads.

3.11.2 Defects on washer faces. Washer faces shall not possess seams exceeding the limits shown in Table IV. Bolts shall not possess mutilation of the washer face sufficient to prevent firm seating.

3.12 Bolt head shank junction. Only longitudinal seams of the depths permitted in Table IV shall be permitted. Slight longitudinal or transverse tool marks or undercuts of depth not to exceed the limits in Table V will be permitted, provided they fair into the shank with no sharp scratches, gouges, or corners.



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TABLE V. Limits for Depth of Tool Marks or Undercuts.

| Bolt size (inches) | Depth of tool marks or undercuts (inches, max) |
|---|---|
| Up to 3/8 incl. | 0.003 |
| Over 3/8 to 5/8 incl. | .004 |
| Over 5/8 to 7/8 incl. | .005 |
| Over 7/8 | .006 |

3.13 Threads. Threads of all screws shall conform to MIL-S-7742 or MIL-S-8879, as specified, and shall be of the sizes specified on the applicable standard. Unless specified, threads shall be right-hand. Threads shall be formed by rolling.

3.13.1 Length of thread. The length of the thread shall be the shank length minus the grip length, as specified on the applicable standard drawings. The grip length of bolts shall be measured from the underside of the head to the end of the full cylindrical portion of the shank.

* 3.14 Dimensions. After plating and finishing, the dimensions shall conform to those specified in the following standards: AN3 thru AN20, AN21 thru AN37, AN42 thru AN49, AN173 thru AN186, MS20073, and MS20074.

3.15 Finish. Noncorrosion-resistant steel bolts shall be cadmium plated in accordance with QQ-P-416, type II, class 2. When plating is specified, the entire bolt, including the threaded portion, shall be plated, unless otherwise specified on the applicable standard drawings. Aluminum alloy bolts shall be anodized in accordance with MIL-A-8625 or chemically surface treated in accordance with MIL-C-5541. Corrosion-resistant steel bolts shall be passivated.

3.15.1 Passivating process for corrosion-resistant steel bolts. This process shall be the final manufacturing operation and consists of immersing the bolts for a period of 20 minutes in the following manner:

a. In a solution containing 15 percent (by weight) of nitric acid at room temperature followed by a thorough rinse in hot water, or

b. In a solution of 6 to 8 percent (by weight) of nitric acid plus 1 to 2 percent (by weight) of hydrofluoric acid heated to a temperature of 120° to 130°F (49° to 54° Centigrade (C)) followed by a thorough rinse in hot water.

3.16 Identification of product. When required, bolts shall be marked for identification purposes as shown on the applicable standard drawing. It is permissible for the bolt manufacturer to include his identification marking on the bolt heads.



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3.17 Workmanship. Bolts shall be made in conformance with high-grade bolt manufacturing practice.

3.18 Surface roughness. Unless otherwise specified, the shank of all close tolerance bolts will have a maximum surface roughness of 63 microinches; the shanks of all other bolts will have a maximum surface roughness of 125 microinches.

* 3.19 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of bolts shall be classified as quality conformance inspection.

4.3 Lot. A lot shall consist of finished bolts which are of the same type and size, fabricated by the same process, heat treated in the same manner, and produced as one continuous run or order, or part thereof.

4.4 Sampling.

4.4.1 Sampling for visual and dimensional attributes. Sampling for visual and dimensionsl attributes shall be at random in accordance with MIL-STD-105. The Acceptance Quality Levels (AQL's) for major, minor A, and minor B classes of characteristics shall be as follows:

a. Major 2.5 percent

b. Minor A 6.5 percent

c. Minor B 10 percent

The total defective items for all characteristics of a given class (e.g., major, minor A, minor B) shall be combined in order to apply the acceptance and rejection provisions of MIL-STD-105.

4.4.1.1 Definitions of defect classes. Major and minor defects are defined in MIL-STD-105. Minor defects are broken down into two classes, minor A and minor B. Definitions for minor A and minor B defects are as follows:



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4.4.1.1.1 Minor A. A minor A defect is classified as a defect which has a slight effect on usability.

4.4.1.1.2 Minor B. A minor B defect is classified as a defect which has no effect on usability but which does not conform to workmanship standards.

4.4.1.2 Classification of defects. All dimensional characteristics are considered defective when out of tolerance.

4.4.1.2.1 AN3 thru AN20 - aircraft and AN173 thru AN186 bolt - aircraft, close tolerance. The classification of defects for AN3 thru AN20 and AN173 thru AN186 bolts shall be as follows:

Critical:

None

Major:

101 Thread size and form (see 4.5.1)
102 Shank diameter, oversize (unthreaded portion)
103 More than two imperfect threads
104 Grip length
105 Fillet under head
106 Cotter pin hole location (where required)
107 Hole in head missing (where required)
108 Dimension across flats
109 Straightness of shank
110 Angle of bearing surface to shank
111 Plating (where required) (visual inspection)
112 Identification
113 Burrs in cotter pin hole or on washer face
114 Microinch finish of shank above maximum tolerances specified in 3.18

Minor A:

201 Length
202 Head thickness
203 Shank diameter, undersize (unthreaded portion)
204 Hole in head, diameter and location (where required)
205 Cotter pin hole diameter (where required)



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Minor B:

301 30-degree chamfer on head
302 45-degree chamfer on thread end
303 Eccentricity of head to shank
304 Burrs, general
305 Washer face diameter and thickness

* 4.4.1.2.2 AN21 thru AN37 bolt and clevis. The classification of defects for AN21 thru AN37 bolts shall be as follows:

Critical

None

Major:

101 Thread size and form (see 4.5.1)
102 Shank diameter (unthreaded portion)
103 Grip length
104 Fillet under head
105 Slot missing
106 Cotter pin hole location (where required)
107 Straightness of shank
108 Angle of bearing surface to shank
109 Plating (where required) (visual inspection)
110 Identification
111 Burrs in cotter pin hole or under head

Minor A:

201 Length
202 Head diameter
203 Slot dimensions
204 Flat on OD of head
205 Head thickness
206 Cotter pin hole diameter (where required)
207 Thread undercut dimensions

Minor B:

301 45-degree chamfer on thread end
302 Eccentricity of head to shank
303 Burrs, general



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4.4.1.2.3 AN42 thru AN49 bolt - eye. The classification of defects for AN42 thru AN49 bolts shall be as follows:

Critical:

None

Major:

101 Thread size and form (see 4.5.1)
102 Shank diameter, oversize (unthreaded portion)
103 More than two imperfect threads
104 Grip length
105 Fillet under head
106 Cotter pin hole location
107 Eye hole diameter and location
108 Diameter of flange
109 Length of tongue and flange
110 Tongue end radius
111 Thickness of tongue
112 Radius tongue to flange
113 Straightness of shank
114 Angle of bearing surface to shank
115 Surface finish, plating (visual inspection)
116 Identification
117 Burrs

Minor A:

201 Length
202 Shank diameter undersize
203 Eccentricity of head to shank
204 Cotter pin hole diameter
205 Flange thickness

Minor B:

301 Chamfer, thread end



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4.4.1.2.4 MS20073 and MS20074 bolt, machine, aircraft, drilled head. The classification of defects for MS20073 and MS20074 bolts shall be as follows:

Critical

None

Major:

101 Thread size and form (see 4.5.1)
102 Shank diameter, oversize (unthreaded portion)
103 More than two imperfect threads
104 Grip length
105 Fillet under head
106 One or more radial holes in head missing
107 Dimension across flats
108 Straightness of shank
109 Angle of bearing surface to shank
110 Plating (where required) (visual inspection)
111 Identification
112 Burrs in cotter pin hole or on washer face
113 Radial holes in head, diameter and location
114 Axial hole in head, diameter, depth, and location

Minor A:

201 Length
202 Shank diameters, undersize (unthreaded portion)
203 Head thickness

Minor B:

301 30-degree chamfer on head
302 45-degree chamfer on thread end
303 Eccentricity of head to shank
304 Burrs, general
305 Washer face diameter and thickness

4.4.2 Sampling for tension, shear, and hardness tests. For each of the tension, shear, and hardness test a separate sample shall be randomly selected in accordance with MIL-STD-414; inspection level IV, with an AQL of 1.5 percent, using Table C-1. The alternate attribute sampling plan shall be in accordance with MIL-STD-105, inspection level I, with an AQL of 1.5 percent. In the case of resubmitted lots over 100,000 in size use the next highest sample size than that used on original inspection.



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4.4.3 Sampling for head structure, bend, and decarburization tests. Sampling shall be in accordance with MIL-STD-105, inspection level S-3, acceptance number zero. For resubmitted lots over 100,000 in size use the next largest sample size than that used in original inspection.

4.4.4 Sampling for magnetic particle inspection. Sampling for magnetic particle inspection shall be at random in accordance with MIL-STD-105, with an AQL of 1.5 percent. The acceptance and rejection numbers of this sampling plan shall apply to those bolts which are judged defective or in nonconformance with the requirements contained herein after microexamination has been performed on those samples revealing magnetic particle indications. Magnetic particle inspection shall not be required for bolts less than 1/4 inch in diameter.

4.4.5 Sampling for penetrant inspection. Sampling for penetrant inspection shall be at random in accordance with MIL-STD-105, with an AQL of 1.5 percent. The acceptance and rejection numbers of this sampling plan shall apply to those bolts which are judged defective or in nonconformance with the requirements contained herein after microexamination has been performed on those samples revealing penetrant indications. Penetrant inspection shall not be required for bolts less than 1/4 inch in diameter.

4.5 Inspections.

* 4.5.1 Dimensions. Dimensions shall be measured by gages or measuring instruments calibrated in accordance with MIL-STD-45662. In case of controversy, gages and instruments from recently certified government laboratories shall be employed. Screw threads shall be checked as specified in MIL-S-7742. Consideration should be given to the possibility of minor differences between various gages which could result in borderline discrepancies.

4.5.2 Tension.

4.5.2.1 Hexagon-head bolts. Hexagon-head bolts shall be tested in accordance with the applicable requirements of FED-STD-151 in tension between the head of the bolt and an AN315 nut of the corresponding size, heat treated to 180,000 pounds per square inch (psi).

4.5.2.1.1 The acceptability of noncorrosion-resistant steel bolts having a grip less than the diameter of the bolt shank shall be based on the hardness test.

4.5.2.1.2 The acceptability of corrosion-resistant steel bolts and aluminum alloy bolts having a grip less than the diameter of the bolt shank shall be based on a test coupon or the same material, diameter, and heat treatment.

4.5.2.2 Eye bolts. Eye bolts shall be tested in accordance with the applicable requirements of FED-STD-151 in tension between the eye of the bolt and a nut of sufficient thickness to develop the full strength of the bolt without stripping the threads.

4.5.2.3 Clevis bolts. Clevis bolts shall be tested in accordance with the applicable requirements of FED-STD-151 in tension between the head of the bolt and an AN320 or MS21083 nut of the corresponding size.



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4.5.3 Shear. The shear test shall be performed in a jig conforming substantially to that shown in Figure 2. Other types of shearing jigs may be used if acceptable to the procuring activity. The acceptability of the bolts which are of insufficient length in the unthreaded portion to conduct the shear test shall be based on the hardness test.

4.5.4 Bend. Bolts shall be bent cold in the unthreaded portion through an angle of 180 degrees over a diameter equal to the bolt diameter for steel bolts and equal to six times the bolt diameter for aluminum alloy bolts. The bending shall be accomplished by the gradual application of pressure in a suitable fixture. The acceptability of the bolts which are of insufficient length in the unthreaded portion to conduct the bend test shall be based on the hardness test.

4.5.5 Hardness. Rockwell hardness readings shall be taken on a smooth, flat, prepared surface at any location on the bolt.

4.5.6 Decarburization. Decarburization shall be determined by microexamination. Specimens shall be taken from a transverse section of the shank of the finished bolt. The etchant shall be 5 percent nital. Microscopic examination shall be made at a magnification of 100 diameters.

4.5.7 Head structure. A longitudinal macrosection of the head, not less than 1/4 inch from the upper portion of shank, after the head-forming and heat treating process, shall be examined for internal structure.

4.5.7.1 Steel bolts. Steel bolts shall be etched in an aqueous solution containing 50 percent (by volume) of hydrochloric acid held at 160° to 180°F (71° to 82°C) for a time sufficient to reveal the macrostructure properly.

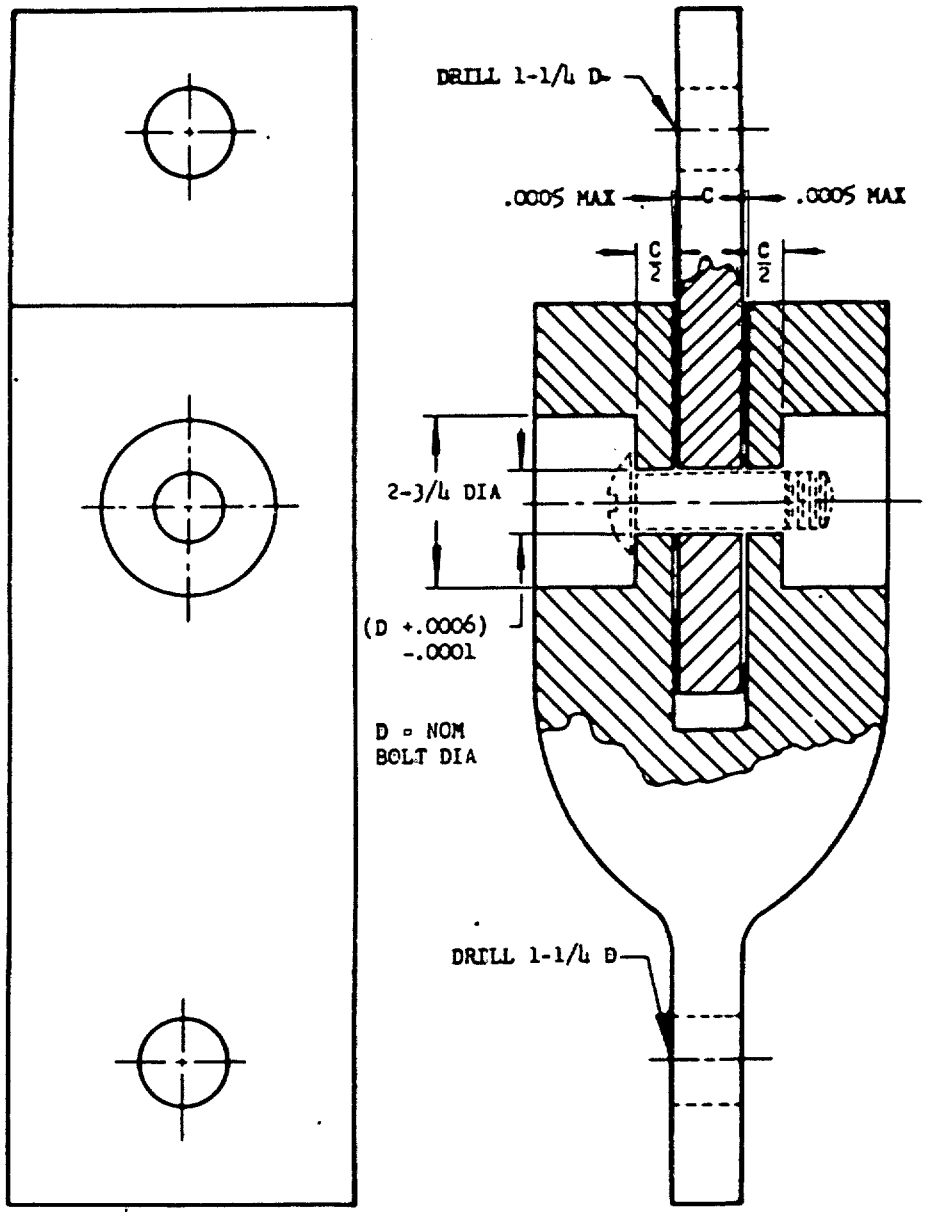
4.5.7.2 Aluminum alloy bolts. Aluminum alloy bolts shall be etched in a 5- to 15-percent (by weight) aqueous caustic soda (NaOH) solution at approximately 180°F (82°C) for a time sufficient to reveal the macrostructure properly. This shall be followed by washing in water and dipping in a 10-percent (by weight) nitric acid solution to remove black stains. Water washing and drying shall follow.

4.5.8 Magnetic particle inspection. The indication of discontinuities in ferromagnetic bolts, such as cracks, seams, and inclusions, shall be determined by magnetic particle inspection. Magnetic particle indications of themselves shall not be cause for rejection. If, in the opinion of the inspector, the indications may be cause for rejection, representative samples shall be taken from those bolts showing indications which shall be further examined by microexamination to determine whether the indicated discontinuities are in accordance with the limits specified herein.

4.5.8.1 Method. Magnetic particle inspection shall be performed in accordance with MIL-I-6868. Such inspection shall, in general, be performed on finished bolts, but in any case subsequent to any processing operation which could adversely affect the part. The magnetizing field shall be parallel to the longitudinal axis of the bolt, primarily for the indication of transverse defects. Bolts shall not be dyed as an indication of magnetic particle inspection in accordance with the sampling requirements of this specification. Personnel conducting magnetic particle inspection shall be certified in accordance with MIL-STD-410.



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| BOLT SIZE | C |
|---|---|
| Under 7/16 | 1/4 |
| 7/16 to 5/8, incl | 1/2 |
| 3/4 to 1, incl | 13/16 |
| 1-1/8 and 1-1/4 | 1-3/16 |

DIMENSIONS IN INCHES.

FIGURE 2. Double shear jig for clevis bolts.



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4.5.9 Penetrant inspection. The indication of discontinuities in aluminum alloy bolts, such as cracks, seams, and inclusions open to the surface shall be determined by penetrant inspection. Penetrant indications of themselves shall not be cause for rejection. If, in the opinion of the inspection, the indications may be cause for rejection, representative samples shall be taken from those bolts showing indications which shall be further examined by micro-examination to determine whether the indicated discontinuities are in accordance with the limits specified herein.

4.5.9.1 Method. Penetrant inspection shall be performed in accordance with MIL-I-6866. Such inspection shall in general be performed on finished bolts, but in any case subsequent to any processing operations which could adversely affect the part. Bolts shall not be dyed as an indication of penetrant inspection in accordance with the sampling requirements of this specification. Personnel conducting penetrant inspection shall be certified in accordance with MIL-STD-410.

4.6 Resubmitted lots. Paragraph 6.4 of MIL-STD-105 shall apply, except that a resubmitted lot shall be inspected by the contractor under supervision of the Government inspector using tightened inspection. For attribute plans of MIL-STD-105, where the original acceptance number was zero, a sample size represented by the next higher sample size code letter shall be chosen with the acceptance number remaining zero. Before a lot is resubmitted, full particulars concerning the cause of previous rejection and the action taken to correct the defects found in the lot shall be furnished by the contractor to the Government inspector.

4.7 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of PPP-H-1581 and section 5 of this specification.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A, C or Commercial as specified (6.2.1).

* 5.1.1 Level A. The bolts shall be preserved in accordance with PPP-H-1581.

* 5.1.2 Level C. The level C preservation for the bolts shall conform to the MIL-STD-794 requirements for this level.

* 5.1.3 Commercial. The commercial preservation of the bolts shall be in accordance with the requirements of ASTM-D3951-82.

* 5.2 Packing. Packing shall be level A, B, C or commercial, as specified (see 6.2.1).

* 5.2.1 Level A and B. The preserved bolts shall be packed for shipment in accordance with PPP-H-1581, except fiberboard boxes shall not be used for level A packing.



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* 5.2.2. Level C. The level C packing for the preserved bolts shall conform to the MIL-STD-794 requirements of this level.

* 5.2.3. Commercial. The preserved bolts shall be packed in accordance with the requirements of ASTM-D3951-82.

* 5.2.4. Palletization. When specified (see 6.2) unitized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Unitized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.3. Marking.

* 5.3.1. Levels A, B, and C. In addition to any special or other identification marking required by the contract (see 6.2.1), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

* 5.3.2. Commercial. Commercial markings shall be in accordance with the requirements of ASTM-D3951-82.

* 5.4. General.

* 5.4.1. Exterior containers. Exterior containers (see 5.2.1, 5.2.2, 5.2.3, and 5.2.4) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

* 5.4.2. Packaging Inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES.

6.1. Intended use. The bolts covered by this specification are intended for use in the manufacture of aircraft and aircraft accessories.

6.1.1. Noncorrosion resistant steel conforming to MIL-S-6050 and AMS6300 may not be suitable for bolts larger than 1/2 inch in diameter.

6.2. Ordering Data.

* 6.2.1. Acquisition requirements. Acquisition documents should specify the following:

a. Title, Number, and date of this specification.

b. Part number of the bolt desired (type, material, thread, lengths).

c. Quantity.

d. Selection of applicable levels of preservation, packaging, marking and commercial/industrial (see 5.1, 5.2 and 5.3).

e. When palletized loads are required (see 5.2.4).



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6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - AR
Navy - AS
Air Force - 99

Review activities:
Army AV, MI
Air Force - 11
Navy - OS
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Preparing activity:
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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-B-6812D | Bolts, Aircraft |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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